

Project context

I'm writing a Bachelor's thesis (TFG) in Audiovisual Systems Engineering at **Universitat Pompeu Fabra (UPF), Barcelona**. The work is on real-time voice-to-instrument audio resynthesis using BRAVE (a streaming Variational Autoencoder, extension of RAVE). I deploy the trained models in Pure Data via the nn~ external for sub-10 ms latency.

Two case studies:

- **Drums** (working prototype that responds to voice/beatbox)
- **Guitar** (working autoencoder for in-distribution input; documented failure for voice-to-guitar)

The thesis must follow **UPF's official TFG template** (Times New Roman, single line spacing, 30-page limit for body, APA citations).

Sources I'm uploading

Before drafting anything, please read and absorb **all** of these:

Reference papers (peer-reviewed sources):

- All papers from my reference list (uploaded as PDFs).
- Pay particular attention to: RAVE (Caillon & Esling 2021), BRAVE (Caspé et al. 2025), DDSP (Engel et al. 2020), NSynth (Engel et al. 2017), TimbreTron (Huang et al. 2018), AutoVC (Qian et al. 2019), HiFi-GAN (Kong et al. 2020), VAE original (Kingma & Welling 2013), GAN original (Goodfellow et al. 2014).

Project documentation (my own notes from the experiments):

- setup_notes.md — installation, patches, training history, noise-floor analysis, configuration details
- workplan.md — week-by-week project plan and current status
- state_of_the_art.md — partial draft of Chapter 2
- All .gin configuration files

Audio + numerical results:

- Quantitative metrics from each training iteration (encoder responsiveness, output amplitude, KL, pred_real/pred_fake)
- TensorBoard screenshots
- Reconstruction WAV files for the audio appendix

Writing rules — apply to every section you draft

Style

1. **Tone:** academic but readable. Engineering thesis style, not literary. First-person plural ("we") is acceptable; first-person singular is also OK at UPF.
2. **Language:** English (UPF accepts English TFGs). Spanish or Catalan only on request.
3. **Tense:** past tense for what was done, present tense for what is true in general.
4. **Paragraphs:** 3–6 sentences each. Avoid one-sentence paragraphs.
5. **No bullet-point dumps** in body text. Use bullets sparingly, only for genuinely enumerable items.
6. **Figures and tables** must be referenced in the text before they appear. Caption every figure and table.

Citations

1. **APA 7th edition** format throughout.
2. **Cite every claim** that is not your own original observation. Examples:
 - "RAVE is a two-stage variational autoencoder (Caillon & Esling, 2021)."
 - "Multi-period discriminators were introduced for vocoders (Kong et al., 2020)."
3. **First mention** of a system or concept: spell it out + cite. **Subsequent mentions:** abbreviation only.
4. **Direct quotation** is rare; paraphrase and cite.
5. **Never invent citations.** If you don't have a source for a claim, mark it [CITATION NEEDED] and I'll find one.
6. **Reference list at the end:** APA format, alphabetical. Use the master reference list I've uploaded.

Structure of every section

1. **Opening sentence** stating what the section is about.
2. **Body paragraphs** developing the argument with citations.
3. **Transition** to the next section if natural.
4. No **"In this section we..."** style meta-commentary unless absolutely needed.

Missing references protocol:

If you need to make a claim that requires a citation but cannot find a suitable source in the materials I uploaded, do NOT invent a citation and do NOT silently omit the claim. Instead:

1. Write the claim in the draft.
2. Immediately after it, add: [CITATION NEEDED: <what kind of source would support this>] — for example: [CITATION NEEDED: a 2022+ paper showing voice-to-instrument style transfer with VAEs].
3. At the end of the chapter draft, list **all** the [CITATION NEEDED] markers together as a numbered list so I can find and resolve them. I will then upload the missing reference, and you can re-draft the paragraph with the proper citation.

What NOT to do

- Don't invent results. Use only the metrics and outcomes I provided.
- Don't pad. UPF has a 30-page hard limit; every paragraph must earn its place.
- Don't use marketing language ("revolutionary", "state-of-the-art" except where literally true and cited).
- Don't speculate about what RAVE/BRAVE/anything "probably" does — cite the source or omit the claim.

Per-chapter instructions

Chapter 1: Introducció (target 2–3 pages)

Sections:

- **1.1 Motivation** — why voice-to-instrument resynthesis matters for live performance; gap in existing tools.
- **1.2 Goals and scope** — concrete deliverables (real-time PD prototype, two case studies, diagnostic methodology).
- **1.3 Contributions** — bullet-style list of 4–5 concrete contributions of the work.

- **1.4 Document structure** — one paragraph mapping the rest of the thesis.

Chapter 2: Estat de l'art / State of the Art (target 6–7 pages)

Already 80% drafted in state_of_the_art.md. Polish it. Add the missing references when I upload the verified APA list. Sections:

- **2.1 Neural audio synthesis** (NSynth, DDSP, RAVE, BRAVE)
- **2.2 Timbre transfer and voice conversion** (TimbreTron, AutoVC, cross-modal challenges)
- **2.3 Datasets** (GuitarSet, Guitar-TECHS, IDMT-SMT-Guitar, Groove MIDI)
- **2.4 Research gap and motivation**

Chapter 3: Metodologia (target 5–6 pages)

Sections:

- **3.1 System architecture** — block diagram (I'll provide), description of pipeline.
- **3.2 Training paradigm** — Phase 1 reconstruction + Phase 2 GAN; beta schedule; latent size selection.
- **3.3 Datasets and preprocessing** — sources, durations, preprocessing pipeline, noise-floor analysis table.
- **3.4 Diagnostic methodology** — TensorBoard metrics + Python diagnostic tests (encoder, decoder, reconstruction).

Chapter 4: Implementació (target 3–4 pages, optional — can merge into Ch. 3)

Sections:

- **4.1 Environment setup** — hardware, software, compatibility patches.
- **4.2 Training pipeline** — config files, auto-resume bat scripts, checkpointing strategy.
- **4.3 Real-time deployment** — model export, Pure Data patches, measured latency.

Chapter 5: Resultats i avaluació (target 8–10 pages — the longest)

Sections:

- **5.1 Working prototypes** — drums (with metrics), guitar autoencoder (with metrics).
- **5.2 Failure taxonomy across six guitar iterations** — one subsection per version, each with: configuration, observation, diagnostic, lesson.
- **5.3 Voice input as out-of-distribution** — quantitative explanation of why voice-to-guitar fails.
- **5.4 Real-time performance** — measured latency.

Chapter 6: Conclusions i treball futur (target 2–3 pages)

Sections:

- **6.1 Summary of contributions**
- **6.2 Limitations**
- **6.3 Future work** (voice preprocessing, solo-only training, larger datasets, alternative architectures)
- **6.4 Closing remarks**

Process I want you to follow

1. **Before drafting anything**, confirm you've read all uploaded sources. Reply with a one-sentence summary of each source.

2. **Ask me which chapter to draft first.** Default to Chapter 1 if no preference.
3. **Draft one chapter at a time.** Don't write the whole thesis in one go — I want to review iteratively.
4. **At the end of each chapter draft,** list the references you used (so I can verify them) and any places you marked [CITATION NEEDED].
5. **Stay within page targets.** If a section seems likely to exceed its target, tell me and propose what to cut.

Iteration protocol

When I send feedback:

- "Tighten X" → reduce wordiness without removing content.
- "Expand X" → add 1–2 more paragraphs with citations.
- "Cite X" → find the strongest citation for a specific claim.
- "Rewrite X formally/informally" → adjust tone.
- Never make changes I didn't ask for.